Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Energy systems with high energy densities for mobile applications using low-pressure and ambient-temperature hydrogen storage*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen (), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Energie & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding/ Marine Systems	*Hull/Deck and Propulsion System Components*	*New & Renewable Energy*	*Hydrogen*
Concept of Proposed Technology	*The "Powerpaste" is a highly viscous magnesium hydride-based energy carrier which enables the release of hydrogen on demand using hydrolysis reaction. Within a reactor vessel operated at low pressure (below 10 bar) the paste is mixed with water. Hydrogen is released which can then be used in a fuel cell for electricity generation or a combustion engine (1-5).* *In comparison to other storage technologies the main advantage lies in the handling and transportation since the paste can be stored at ambient-temperature and no high-pressure equipment is needed which improves the safety of operation. Furthermore, "Powerpaste" excels with high gravimetric electrical energy densities above 1500 Wh/kg which makes it an ideal candidate for energy carriers in mobile applications.* *The main use case is seen in applications with long operation duration and medium power output. Past mobile*			

	developments were focused on small vehicles and drones around 1kW, although maritime applications or systems with higher power output, e.g. 5 kW-10 kW, are also conceivable.
Rationale for Joint R&D Project	*Although the underlying hydrolysis-based metal-hydride slurry technology is already technically mature, deploying it in real-world, application-specific systems, especially for mobile and off-grid use, still requires substantial R&D. Key work packages include system integration (reactor, dosing, heat and water management, controls), scaling to relevant power levels, packaging and weight optimization, safety engineering, reliability testing, and validation under representative operating conditions. These activities require significant upfront investment and cannot be completed through laboratory development alone.* *Collaboration is essential because no single partner can efficiently cover the full development chain from materials handling to a certified, field-ready system. The project must combine complementary expertise and infrastructure: slurry formulation and IP know-how, system and mechanical engineering for robust mobile integration, electronics/control development, safety and regulatory competence, and access to realistic test environments and end-user requirements. Joint R&D reduces technical and commercial risk, accelerates iteration cycles, and ensures that the patented concepts (6-7) are validated in operationally relevant settings.* *From a market perspective, on-demand hydrogen generation still lacks standardized products and established supply chains, which limits private investment despite proven technical feasibility. Public co-funding helps de-risk the adaptation to new applications and supports the creation of validated reference systems that can be transferred into scalable products.*
Key Words	*1. Hydrogen-based energy systems* *2. Hydrogen-on-demand generation* *3. low-pressure components* *4. Ambient temperature storage* *5. High energy densities*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (7)
R&D Objectives	*The objective of the R&D is to develop a hydrogen-based energy systems which supplies a mobile application with electrical energy and stores the needed chemical energy in the MgH_2-based slurry (Powerpaste). The energy system will be designed in close cooperation with the Korean partner to enable a swift integration into one of the Korean partner's systems. The goal is refrain from using any high-pressure components and reaching high energy densities on a system level.*			
R&D Target Specifications	• *Development of a full-scale prototype/commercial product for maritime or aviation applications powered by hydrogen* • *Reaching high energy densities on a system-level (400-700 Wh/kg depending on the application)* • *Refraining from using any high-pressure or cryogenic equipment and therefore safer/easier handling and fuel transportation*			
R&D Plan	*Depending on the final application scenario the R&D Plan may vary. The presented plan is exemplary:* *<u>Phase I (Pre-study)</u> — Requirements, feasibility, and R&D blueprint (Month 1–6)* *Task 1 – Use-case definition & requirements engineering* • *Lead: Korean SME* • *Outputs: requirements baseline (use cases, constraints) and key performance indicators (KPIs) defined, acceptance, risk and initial test concept evaluated* *Task 2 –Feasibility study, architecture & validation plan* • *Lead: Fraunhofer IFAM* • *Outputs: pre-study report, feasibility confirmed, system architecture & interfaces defined, updated R&D roadmap for Phase II* *<u>Phase II (Main R&D)</u> – Technology development, prototyping, Korea testing, and iteration (Month 7–36)* *Task 3 – Core technology development & lab proof-of-concept (Month 7–14)* • *Lead: Fraunhofer IFAM* • *Methodologies: technology development, controlled benchmarking vs. KPIs, robustness and sensitivity checks* • *Outputs: validated core technology modules, performance baseline report, updated risk register* *Task 4 – Prototype build & integration (Month 12–18)* • *Lead: Fraunhofer IFAM (continuous alignment with SME)* • *Methodologies: modular prototyping, interface-based integration, engineering documentation and configuration management* • *Outputs: Prototype 1, interface and integration documentation, deployment package for Korea testing* *Task 5 – Lab pre-validation & field-test readiness (Month 16–18)* • *Lead: Fraunhofer IFAM + SME review* • *Methodologies: standardized test protocols, reproducibility checks* • *Outputs: pre-validation report, "field-test readiness" package, finalized on-site test procedure* *Task 6 – On-site testing and validation in Korea (Round 1) (Month 19–22)* • *Lead: Korean SME (with Fraunhofer IFAM support)* • *Methodologies: field trial execution, operational data collection, structured defect/performance logging, user feedback capture* • *Outputs: validation report vs. KPIs, prioritized improvement backlog* *Task 7 – Iteration, optimization & prototype refinement (Month 22–28)* • *Lead: Fraunhofer IFAM (with SME feedback and re-test support)*			

	• *Methodologies: root-cause analysis, redesign/parameter tuning, regression testing, reliability improvements* • *Outputs: Prototype 2, updated documentation, readiness for re-validation* *Task 8 – On-site testing and validation in Korea (Round 2 / extended pilot) (Month 28–32)* • *Lead: Korean SME* • *Methodologies: extended pilot testing under broader operating conditions, stability/reliability evaluation, acceptance testing* • *Outputs: pilot validation, final gap analysis (if any), go/no-go input for final demonstrator* *Task 9 – Final demonstrator, consolidation & exploitation package (Month 32–36)* • *Lead: Fraunhofer IFAM + Korean SME* • *Methodologies: final optimization and verification, documentation* • *Outputs: final demonstrator, final KPI compliance report, technical dossier (specs, interfaces, test results), commercialization input* *Expected milestones* • *M1 (End of Month 3): Requirements baseline, Key performance factors (KPI) and acceptance criteria approved (SME-led sign-off)* • *M2 (End of Month 6): Pre-study completed: feasibility confirmed, architecture/interfaces and Phase II roadmap finalized* • *M3 (End of Month 14): Core technology reaches lab KPI thresholds (proof-of-concept validated)* • *M4 (End of Month 18): Prototype 1 completed and lab pre-validation passed (field-test ready)* • *M5 (End of Month 22): Korea on-site validation Round 1 completed; improvement backlog agreed* • *M6 (End of Month 28): Prototype 2 completed with regression testing and readiness for extended pilot* • *M7 (End of Month 32): Korea pilot/validation Round 2 completed; final acceptance assessment available* • *M8 (End of Month 36): Final demonstrator delivered with consolidated documentation and exploitation package*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will co-lead the technology development with Korean SME and provide the core scientific/technical contributions. Based on the SME's boundary conditions and market needs, it will define the R&D roadmap, agree on measurable KPIs and test/acceptance criteria, and translate them into a clear system specification. In close alignment with the Korean SME, Fraunhofer IFAM will design and build a functional prototype, including interface definitions and technical documentation. It will provide the required research resources (experts, lab/tools) and perform pre-validation in a controlled environment before shipment. During the on-site testing in Korea, Fraunhofer IFAM will support analysis of results, iterate on the design, and deliver consolidated technical reports and (as agreed) IP/documentation inputs.*
Expected R&R of Korean SME Partner	*The Korean SME will lead the market and application perspective by providing boundary conditions and defining what is needed for real-world adoption. It will specify use cases, requirements, and constraints (e.g., cost, performance, reliability, form factor, compliance/standards) and translate these into acceptance criteria. The SME will collaborate closely with the Fraunhofer IFAM on specifications, priorities, and integration conditions, and provide relevant operational scenarios, data, and interface requirements. It will set up and run on-site testing in Korea (test environment, QA process, pilot/field trial), collect structured feedback, and document issues and performance gaps. Together with Fraunhofer IFAM, it will drive iterative improvements and support commercialization planning and exploitation/IP strategy in line with consortium rules.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project aims to deliver a high-TRL (up to TRL 8) hydrogen-based energy system for mobile applications in aviation and/or maritime applications which is able to convert the chemical energy of Powerpaste into electrical energy using hydrolysis.* *The project scope can also include integration of the hydrogen source (generator employing PowerPaste technology) with other systems (for example related to core competence of the Korean SME) for example: integration of Hydrogen Generator with Fuel Cell on the board of flying drone. MgH_2-based hydrolysis technology has not been employed so far in any Korea sector.* *Technologically, it will establish a technology which is able to deliver hydrogen on demand using no high-pressure and cryogenic components and there enabling safer operation. Economically, it will reduce infrastructure, transport, and safety costs in comparison to high-pressure and cryogenic hydrogen-storage. The market share will depend on the final application scenario. The project will create high-skill jobs in materials science, chemical engineering, system manufacturing, and energy integration.*

6. References

(1) Tegel, M.; Schöne, M.; Kieback, B.; Röntzsch, L. "An efficient hydrolysis of MgH_2-based materials." International Journal of Hydrogen Energy, 42(1), 183–191 (2017). DOI: 10.1016/j.ijhydene.2016.09.084

(2) Fraunhofer IFAM. "POWERPASTE FOR OFF-GRID POWER SUPPLY" (White Paper, May 2022). https://www.ifam.fraunhofer.de/content/dam/ifam/en/documents/dd/Infobl%C3%A4tter/White_paper_POWERPASTE_final.pdf

(3) Fraunhofer IFAM (Dresden). "POWERPASTE" technology page (key facts, performance and application context). https://www.ifam.fraunhofer.de/en/Aboutus/Locations/Dresden/HydrogenTechnology/powerpaste.html

(4) Fraunhofer ZESS (Fraunhofer IKTS). "POWERPASTE – chemical hydrogen storage: Hydrolysis" (reaction and key figures). https://www.zess.fraunhofer.de/en/competencies/hydrogen_technologies/powerpaste-hydrolysis.html

(5) Fraunhofer (Mediathek). "Video: POWERPASTE – Wasserstoffantriebe für Kleinfahrzeuge" (official Fraunhofer video page). https://www.fraunhofer.de/de/mediathek/filme/filme-2021/powerpaste.html

(6) European Patent EP3154900B1. "Composite material for hydrolytically generating hydrogen … and possible applications." https://patents.google.com/patent/EP3154900B1/en

(7) US Patent US10239753B2 (PDF). "Composite material for hydrolytically generating hydrogen … and possible applications." https://patentimages.storage.googleapis.com/87/3f/fa/54a7bc14ab736d/US10239753.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization